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Socioeconomic burden of adult attention deficit/hyperactivity disorder in Japan: a nationwide cross-sectional observational study

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Abstract

Background Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental disorder that negatively affects social, academic, and occupational functioning. Although more common in children, ~2.5% of adults worldwide have ADHD. In this cross-sectional observational study, we estimated the annual socioeconomic burden of adults with ADHD in Japan (societal perspective).

Methods We used retrospective data from the JMDC claims database, a prospective web-based survey of adults with self-reported ADHD, adult family members living with an adult with ADHD, and adults without ADHD who had an Adult ADHD Self-Report Scale score < 4 (control population), and Japanese government agency websites to estimate medical costs (outpatient costs and drug costs), social welfare costs (disability pensions and employment support services), and productivity losses due to unemployment, low income, absenteeism, and presenteeism. In addition to the base-case analysis, we conducted scenario and sensitivity analyses to examine the effects of changing estimation and matching methods on costs, as well as a subgroup analysis of costs stratified by the number of psychiatric comorbidities.

Results Cost estimates were based on 30,730 adults with an ADHD diagnosis in the JMDC database and, from the web-based survey, 309 adults with self-reported ADHD, 309 adult family members of adults with ADHD, and 927 adults in the control population. Using an estimated 260,120 adults with ADHD in Japan, the base-case analysis estimated annual medical costs at 74.0 billion Japanese yen (JPY), social welfare costs at 389.6 billion JPY, and productivity losses at 571.1 billion JPY. Productivity losses in family members were estimated at 542.2 billion JPY. Total socioeconomic burden was estimated at 1.58 trillion JPY (~US$10.9 billion) (~4 million JPY [~US$29,000] per adult with ADHD). Social welfare costs and productivity losses increased with the number of comorbidities. In the scenario analyses, cost estimates varied from −68.9% to +96.1% relative to the base case and were most strongly affected by the method used to estimate the total number of adults with ADHD.

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Conclusion This study, the first estimate of the annual socioeconomic burden of adult ADHD in Japan, underscores the necessity of more appropriate support for adult patients with ADHD and their families to reduce productivity losses.

Trial registration University hospital Medical Information Network Center (UMIN000051654; registered July 20, 2023).

Keywords Attention deficit disorder with hyperactivity, Cost of illness, Direct medical costs, Family members, Health care costs, Productivity losses, Social welfare costs, Socioeconomic burden, Unemployment, Work performance

Background

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental disorder characterized by inattention and hyperactivity/impulsivity at levels that negatively affect social, academic, and occupational functioning [1–3]. The worldwide prevalence of ADHD has been estimated to be about 7.2% in children and adolescents [4] and about 2.5% in adults [5], although estimates vary widely depending on criteria used [6, 7]. In Japan, the prevalence of ADHD has been reported to be 5.8–7.7% in children and adolescents [8, 9] and 1.65% in adults [10]. However, these estimates were based on community-level surveys, and no nationwide prevalence estimates are available for adult ADHD in Japan. Because the incidence of newly diagnosed ADHD among Japanese adults increased steadily between 2010 and 2019 [11], the current prevalence may be higher than previously suggested. ADHD is often diagnosed in childhood, and although not all patients continue to meet the full diagnostic criteria for ADHD in adulthood, functional disabilities or subthreshold symptoms persist at a high rate into adulthood [12, 13]. In addition, patients with ADHD frequently have psychiatric comorbidities [14], which place an additional burden on patients.

The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision states that ADHD in adults is associated with poorer occupational performance, attainment, and attendance, and a higher probability of unemployment [1]. Reduced work productivity (absenteeism and presenteeism) in adult patients with ADHD has been reported in various countries, including Japan [15–17]. Productivity is likely reduced by the core symptoms of ADHD, which can adversely affect time management, organization, and interpersonal relationships [18, 19]. Moreover, poor understanding of ADHD by society may lead to stigma [20] and reduce the opportunity for adults with ADHD to obtain and hold a job [21]. Under such circumstances, some people may be reluctant to visit a psychiatrist because they fear being diagnosed with ADHD, resulting in a lack of or a delay in receiving appropriate diagnosis, treatment, or welfare.

In Japan, people with ADHD are eligible to receive medical treatment as well as work support, including social skills training, self-reliance support systems, and/or a disability pension in severe cases [22]. However, it is unclear whether current employment support services are effective enough to help adults with ADHD who are willing to work. In addition, problems associated with caregiving may impose a considerable burden on family members of adults with ADHD [23]. Recently, employers have started to include people with neurodevelopmental disorders in their workplaces under the concept of neurodiversity, which recognizes the competitive advantages of a neurodiverse workforce in rapidly changing and highly specialized work environments [24, 25]. Thus, discussions regarding appropriate work support for adults with ADHD and family members who are willing to work, and its socioeconomic impact, would be very meaningful. In this context, the estimation of the socioeconomic burden of ADHD would provide useful data for policy discussions. Multiple analyses of the socioeconomic burden of pediatric and/or adult ADHD have been conducted, primarily in the USA and Europe [26–28]. However, to the best of our knowledge, there have been no reports estimating the socioeconomic burden of adult ADHD in Japan or in Asia more broadly.

We conducted a cross-sectional observational study to estimate the annual socioeconomic burden of adult patients with ADHD in Japan from a societal perspective by combining a retrospective claims database analysis with a web-based survey and statistics from the Japanese government. We estimated medical costs (ADHD drugs and outpatient healthcare) and costs due to productivity losses and social welfare expenses. In addition, as work productivity and employment of adult family members of adults with ADHD can be impaired [23], we also estimated productivity losses in family members.

Methods

Study design

This was a nationwide cross-sectional observational study using retrospective data from the JMDC claims-based database, prospective data from a web-based survey, and publicly available data from the Japanese government. JMDC (JMDC Inc., Tokyo, Japan) is a nationwide database of receipts and medical examination data from multiple health insurance societies in Japan, which started in 2005. As of February 2022, JMDC included data from approximately 14 million registered

individuals. As data from JMDC were retrospective and anonymous, informed consent was not required.

The web-based survey, conducted between July 26, 2023, and August 9, 2023, was completed by participants from panels managed by Macromill, Inc. (Tokyo, Japan). Participants were recruited from a general panel and a panel of patients with ADHD and family members, and participants were located throughout Japan. Data collection in the general panel was allocated by sex and age category according to the composition of the adult population in Japan. An additional survey, conducted between July 26, 2023, and August 2, 2023, recruited participants from an adult ADHD population who visited Nihon Chouzai pharmacies in the Kanto area (Nihon Chouzai Co., Ltd., Tokyo, Japan) with a prescription for ADHD-specific medication. These participants were considered more likely to have a diagnosis of ADHD compared with participants in the Macromill panels who self-reported ADHD, and survey results were used for validation of the Macromill panel results. Survey participants in the Macromill panels received points equivalent to 60 Japanese yen (JPY) and those from Nihon Chouzai received an Amazon gift card equivalent to 500 JPY as compensation. Survey participants provided informed consent on the web screen before starting the survey and were de-identified by Macromill and Nihon Chouzai prior to providing data to the researchers.

Data on the Japanese population, outpatient and drug costs, public employment support services, unemployment, income, and working hours were derived from the most recent data available from the websites of government agencies, including the Ministry of Health, Labour and Welfare (MHLW), the Statistics Bureau of Japan, the Ministry of Internal Affairs and Communications, and the National Tax Agency JAPAN.

The study was conducted in compliance with ethical principles based on the Ethical Guidelines for Life Science, Medical and Health Research Involving Human Subjects.

The study protocol was approved by the Non-Profit Organization MINS Institutional Review Board (ID: 230222) and registered at the University hospital Medical Information Network Center (UMIN000051654) on July 20, 2023.

Study population

JMDC claims database

Adults (aged ≥ 18 and < 65 years at April 1, 2022) registered in the JMDC database who had a diagnosis of ADHD, based on the International Statistical Classification of Diseases and Related Health Problems 10th Revision code F90, during the Japanese financial year 2022 (April 2022 to March 2023) were eligible for inclusion.

Web-based survey

Male or female adults (aged ≥ 18 and < 65 years at informed consent) who were living in Japan were eligible for inclusion if they met the following criteria for each group:

- Adults with ADHD: self-reported diagnosis of ADHD.
- Adult family members of adults with ADHD: no diagnosis of ADHD; currently living with a family member aged ≥ 18 years who is diagnosed with ADHD.
- Adult ADHD Self-Report Scale (ASRS) negative non-ADHD adults (control population): no diagnosis of ADHD; no family member aged ≥ 18 years who is diagnosed with ADHD; ASRS score < 4.

The ASRS is an 18-item self-administered test used to screen for adult ADHD, which has been validated in Japanese [29]. In the 6-item screening version used here, which is widely used as a screener in Japanese clinical practice [30], the result is considered positive if at least four of six questions in Part A of the test are answered positively [29, 31, 32]. An adult ASRS-positive non-ADHD population was also identified for exploratory analyses not reported herein.

The target sample size for adults with ADHD was set at 300, based on a previous study by the Nomura Research Institute, which recruited more than 300 adults with ADHD from Macromill panels [33]. To provide a similar sample size, the target number of adult family members of adults with ADHD was also set at 300. The target sample size for adult ASRS-negative non-ADHD adults was set at 900 (i.e. three times the size of the ADHD and family member samples) to allow for matching.

Base-case analysis

The analysis methods used to estimate the socioeconomic burden of ADHD in Japan are summarized below; details are shown in Additional file 1 (Supplementary Methods and Fig. S1). The number of adult patients with ADHD in Japan, defined as the number of adult patients with ADHD who received outpatient care from April 2022 to March 2023, was calculated based on data from JMDC and Japanese population data from the Statistics Bureau of Japan. The formula used was as below:

$$\text{The number of adult patients with ADHD} = \sum_{i=18}^{64} \left(n_{\text{adhd,male,i,jmdc}} \times \frac{N_{\text{male,i,japan}}}{n_{\text{male,i,jmdc}}} + n_{\text{adhd,female,i,jmdc}} \times \frac{N_{\text{female,i,japan}}}{n_{\text{female,i,jmdc}}} \right)$$

where

$n_{adhd,male,i,jmdc}$ = The number of adult male outpatients with ADHD at the age of i derived from JMDC;

$n_{male,i,jmdc}$ = The number of adult male individuals at the age of i derived from JMDC;

$N_{male,i,japan}$ = The number of adult male individuals at the age of i in Japan derived from population estimates, Statistics Bureau of Japan;

$n_{adhd,female,i,jmdc}$ = The number of adult female outpatients with ADHD at the age of i derived from JMDC;

$n_{female,i,jmdc}$ = The number of adult female individuals at the age of i derived from JMDC;

$N_{female,i,japan}$ = The number of adult female individuals at the age of i in Japan derived from population estimates, Statistics Bureau of Japan;

i = Age.

Medical costs, defined as the total costs of medical care (outpatient costs and drug costs) for adult patients with ADHD from April 2022 to March 2023, were calculated by multiplying average medical costs per person derived from JMDC data by the number of patients. Drugs for ADHD (in-hospital and pharmacy) were identified based on the World Health Organization's Anatomical Therapeutic Chemical codes N06BA04 (methylphenidate), N06BA09 (atomoxetine), and C02AC02 (guanfacine).

Social welfare expenses, defined as the total annual expenses incurred for adult patients with ADHD as social welfare services provided by the government, were calculated by summing disability pensions and public employment support services. Data for disability pensions were derived from the web-based survey. Data for public employment support services were derived from the web-based survey and from MHLW websites [34, 35].

Productivity losses were defined as the total of losses due to unemployment, low income, absenteeism, and presenteeism. The web-based survey included questions from the Work Productivity and Activity Impairment questionnaire [36], which asked how many hours were missed from work (absenteeism) and to what extent their work productivity was impaired (presenteeism) because of health problems during the previous 7 days. For adult family members, the absenteeism and presenteeism questions referred both to problems due to ADHD in the adult with ADHD and to their health problems. Productivity losses were calculated for the adult ADHD population and for adult family members of adults with ADHD. The number of adult family members was estimated by multiplying the proportion of family members in the web-based survey relative to the total number screened for the survey by the number of adults in the Japanese population using data from the Statistics Bureau of Japan [37]. Prior to analysis, exact matching was performed between each population (adults with ADHD; family members) and ASRS-negative non-ADHD adults based on age, sex, and place of residence (region of Japan) data collected in the web-based survey. Losses due to unemployment and losses due to low income were calculated by taking the difference between the mean values in the Japanese population derived from the websites of the Ministry of Internal Affairs and Communications [38, 39] and the National Tax Agency JAPAN [40], and the mean values in adults with ADHD derived from the survey. Losses due to absenteeism and losses due to presenteeism were calculated by taking the difference between the mean values in adults with ADHD and ASRS-negative adults without ADHD using data from the web-based survey and the MHLW [41].

The total socioeconomic burden (primary endpoint) was the sum of medical costs (adults with ADHD), social welfare expenses (adults with ADHD), and productivity losses (adults with ADHD and adult family members).

Additional analyses

Subgroup analysis based on number of comorbidities

Social welfare expenses and productivity losses were calculated as above with minor modifications (see Additional file 1 [Supplementary Methods]) in adults with ADHD stratified by the number of self-reported psychiatric comorbidities (0, 1, 2, or ≥3) provided by participants in the web-based survey.

Scenario analysis

To evaluate the effect of changes in the estimation methods on costs, we conducted a total of 40 scenario (sensitivity) analyses (10 scenarios for medical costs × four scenarios for productivity losses). Variations in methodology were as follows (method used in the base-case analysis underlined):

- Estimation of number of adults with ADHD and medical costs: top-down approach based on MHLW statistics or bottom-up approach based on web-based survey versus *bottom-up approach based on JMDC*.
- Definition of ADHD: ADHD as main diagnosis only versus *ADHD as main diagnosis and subdisease*.
- Definition of the drugs: ADHD medications plus medications for associated mental disorders versus *ADHD medications only*.
- Estimation of unemployment rate in non-ADHD adults: based on the result of web-based survey versus *statistics from the Ministry of Internal Affairs and Communications*.
- Estimation of average annual income in non-ADHD adults: based on the result of web-based survey versus *statistics from the National Tax Agency JAPAN*.

Details are provided in Additional file 1 (Supplementary Methods).

Sensitivity analysis

As an additional sensitivity analysis, productivity losses in adult patients with ADHD and adult family members of adult patients with ADHD were also estimated after matching with ASRS-negative non-ADHD adults by sex, age, place of residence, educational background, and occupation, and without any matching.

Comparison of Macromill and Nihon Chouzai survey participants

To assess the validity of the web-based survey data, we compared the results from the combined Macromill panels (Macromill general panel and Macromill ADHD patient panel) with those from Nihon Chouzai. Prior to analysis, exact matching was performed between adult patients with ADHD from the Macromill panel and those from Nihon Chouzai based on sex, age, place of residence, educational background, and occupation.

All costs are reported in JPY; during 2022, the average exchange rate was approximately 137.4 JPY to 1 US$. All statistical analyses were conducted using R, Version 4.2.0 or later (R Foundation for Statistical Computing, Vienna, Austria).

Results

Demographic and clinical characteristics

In the JMDC claims database, we identified 30,730 adult outpatients with a diagnosis of ADHD (Fig. 1a). Of these, 17,449 (56.8%) were male and 13,281 (43.2%) were female; the mean (standard deviation; SD) age was 32.2 (11.0) years.

In the web-based survey using the Macromill panels, we identified 309 adults with ADHD and 309 adult family members of adults with ADHD (Fig. 1b). We also identified 927 adults without ADHD who were ASRS negative and were used for matching analyses, as well as 309 adults without ADHD who were ASRS positive (used for exploratory analyses not reported herein). Demographic and clinical characteristics are shown in Table 1 and Additional file 1 (Table S1). Adults with ADHD had psychiatric comorbidities at a higher rate than ASRS-negative

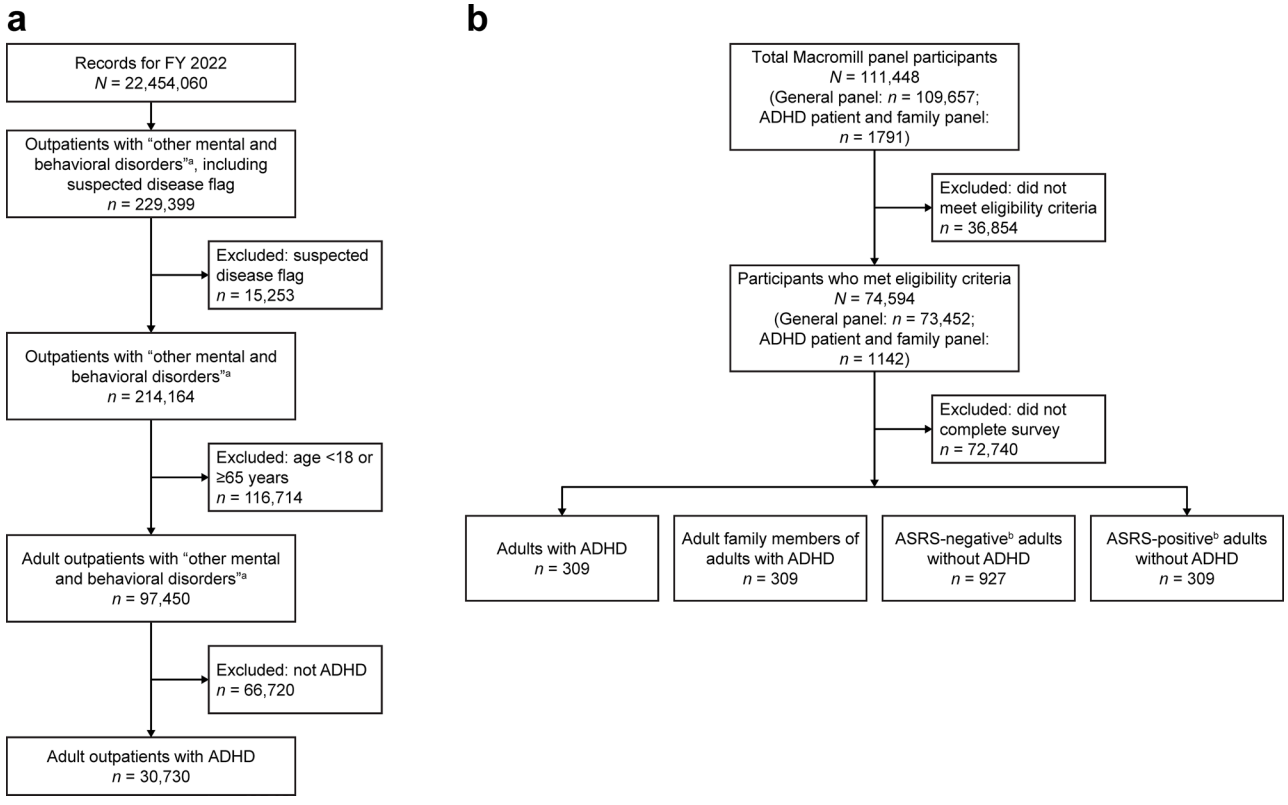


Fig. 1 (**a**) Identification of adults with ADHD in the JMDC claims database in the Japanese 2022 financial year; (**b**) Identification of adults with ADHD, adult family members of adults with ADHD, and ASRS-negative non-ADHD adults in the web survey. [a] ICD-10 codes F02, F04–F09, F49–F69, F80–F99; the code for ADHD is F90. [b] ASRS-positive individuals did not have a diagnosis of ADHD but had an ASRS score ≥ 4; ASRS-negative individuals did not have a diagnosis of ADHD and had an ASRS score < 4. *ADHD* attention deficit/hyperactivity disorder, *ASRS* Adult ADHD Self-Report Scale, *FY 2022* Japanese financial year 2022 (April 2022 to March 2023), *ICD-10* International Statistical Classification of Diseases and Related Health Problems 10th Revision

Table 1 Demographic and clinical characteristics of Japanese adults with ADHD and adult family members in the web-based survey

Variable	Adults with ADHD (*N* = 309[a])	Adult family members (*N* = 309[a])	ASRS-negative non-ADHD adults (*N* = 927)
Sex, *n* (%)			
Male	187 (60.5)	144 (46.6)	468 (50.5)
Female	122 (39.5)	165 (53.4)	459 (49.5)
Age, years, mean (SD)	36.6 (11.0)	42.7 (11.5)	44.2 (11.5)
ASRS score, mean (SD)	3.8 (1.8)	2.0 (1.8)	1.1 (1.1)
Employment status, *n* (%)			
Yes	248 (80.3)	239 (77.3)	747 (80.6)
No	61 (19.7)	70 (22.7)	180 (19.4)
Income[b], 10,000 JPY, mean (SD)	293.3 (284.5)	323.3 (368.7)	341.2 (319.7)
WPAI[b]			
Working hours (past 7 days), mean (SD)	36.0 (20.8)	33.1 (19.1)	37.5 (17.9)
Hours of work missed due to health problems[c] (last 7 days), mean (SD)	5.1 (10.6)	3.7 (10.5)	1.6 (7.3)
Hours of work missed for other reasons (last 7 days), mean (SD)	14.3 (27.1)	13.4 (27.2)	15.6 (30.3)
Degree that health problems[c] affect productivity, %, mean (SD)	43.2 (30.9)	32.3 (30.4)	17.4 (24.3)
Degree that health problems[c] affect regular daily activities, %, mean (SD)	48.8 (31.5)	37.6 (31.5)	22.0 (27.4)
Disability pension benefits, *n* (%)			
Disability basic pension	62 (20.1)	26 (8.4)	17 (1.8)
Disability basic pension + disability welfare pension or disability mutual aid pension	66 (21.4)	45 (14.6)	20 (2.2)
Disability welfare pension or disability mutual aid pension	11 (3.6)	16 (5.2)	10 (1.1)
Disability allowance (lump sum)	13 (4.2)	7 (2.3)	6 (0.6)
None	157 (50.8)	215 (69.6)	874 (94.3)
Employment support, *n* (%)			
Transition support for employment	85 (27.5)	58 (18.8)	19 (2.0)
Continued employment support - type A[d]	62 (20.1)	47 (15.2)	11 (1.2)
Continued employment support - type B[d]	57 (18.4)	51 (16.5)	11 (1.2)
Comorbidities, *n* (%)			
Substance use disorder	14 (4.5)	16 (5.2)	2 (0.2)
Bipolar disorder	36 (11.7)	29 (9.4)	9 (1.0)
Depression	118 (38.2)	57 (18.4)	39 (4.2)
Anxiety disorder	78 (25.2)	53 (17.2)	23 (2.5)
Adjustment disorder	60 (19.4)	48 (15.5)	15 (1.6)
Tic disorder	13 (4.2)	12 (3.9)	2 (0.2)
Insomnia	93 (30.1)	44 (14.2)	40 (4.3)
Learning disability	32 (10.4)	31 (10.0)	2 (0.2)
ASD	108 (35.0)	102 (33.0)	9 (1.0)
Epilepsy	15 (4.9)	25 (8.1)	6 (0.6)
Schizophrenia	30 (9.7)	27 (8.7)	8 (0.9)
None of the above	50 (16.2)	95 (30.7)	814 (87.8)

[a] The adults with ADHD group and the adult family members group were independently recruited

[b] Questions relating to income and the WPAI (certified Japanese translation) were answered only by participants who indicated they were employed

[c] Questions relating to health problems were phrased as "your own health problems" and "problems caused by ADHD in family members living with you"

[d] Type A provides ongoing support for people with disabilities who have difficulty with regular employment but who are able to work under an employment contract, and it guarantees them a minimum wage; type B provides support for people with disabilities who have difficulty with regular and contract-based employment, and it provides payment for production activities undertaken at the recipient's own pace

ADHD attention deficit/hyperactivity disorder, *ASD* autism spectrum disorder, *ASRS* Adult ADHD Self-Report Scale, *JPY* Japanese yen, *SD* standard deviation, *WPAI* Work Productivity and Activity Impairment

non-ADHD adults, most commonly depression, autism spectrum disorder, insomnia, and anxiety disorder.

Adult family members were slightly older than the adults with ADHD (Table 1). Psychiatric comorbidities were generally less common in family members than in adults with ADHD, but more common than in ASRS-negative non-ADHD adults.

Of 60 participants in the Nihon Chouzai panel, 52 completed the web-based survey. After matching, the characteristics of adults with ADHD from the Macromill and

Nihon Chouzai panels were generally similar, although the use of social welfare services was higher in participants from the Macromill panel (Additional file 1 [Table S2]).

Base-case analysis

Medical costs in adult patients with ADHD

The annual medical costs per adult with ADHD in the JMDC database were estimated to be 284,445 JPY, which included 54,807 JPY for outpatient medical expenses and 229,639 JPY for ADHD drug costs (Table 2). Based on an estimated 260,120 adult patients with ADHD in Japan (Additional file 1 [Table S3]), the total medical costs for adult patients with ADHD in Japan were estimated to be approximately 74.0 billion JPY, including approximately 14.3 billion JPY for outpatient costs and approximately 59.7 billion JPY for ADHD drug costs (Table 2).

Social welfare expenses for adult patients with ADHD

In the web-based survey, after matching, 48.7% of patients with ADHD received a disability pension (Additional file 1 [Table S1]), at an average amount of 752,329 JPY per person (Additional file 1 [Table S3]). The total cost of disability pensions for Japanese adults with ADHD was estimated to be approximately 95.2 billion JPY (Table 2). Transition support for employment, continued employment support type A, and continued employment support type B was received by 27.3%, 20.0%, and 18.7% of patients with ADHD, respectively (Additional file 1 [Table S1 and Table S3]). The total cost of public employment support services received by Japanese adults with ADHD was estimated to be approximately 294.4 billion JPY (Table 2). Together, the total social welfare expenses for adult patients with ADHD were estimated to be approximately 389.6 billion JPY.

Productivity losses in adult patients with ADHD

The unemployment rate in adults with ADHD in the web-based survey was 19.0%, whereas the average unemployment rate in the Japanese population was 14.7% (Additional file 1 [Table S3]). Total losses due to unemployment of adults with ADHD in Japan were estimated to be approximately 49.6 billion JPY (Table 2). The average annual income of adults with ADHD in the web-based survey was 2,934,859 JPY, whereas the average annual income of the Japanese population was 4,430,000 JPY (Additional file 1 [Table S3]). Total losses due to low income of adults with ADHD in Japan were estimated to be approximately 315.0 billion JPY. Total losses due to absenteeism and presenteeism in adults with ADHD in Japan were estimated to be approximately 74.2 billion JPY and 132.3 billion JPY, respectively (Table 2). Together, the total productivity losses for adults with ADHD were estimated to be approximately 571.1 billion JPY.

Table 2 Base-case analysis of the total annual socioeconomic burden of adult ADHD in Japan in the Japanese 2022 financial year

Population	Outcomes		Value (JPY)	Percentage of each burden to the total burden of adult ADHD in Japan (%)	Burden per patient/family member (JPY)
Adults with ADHD	Medical costs	Outpatient costs	14,256,285,783	1.38	54,807
		Drug costs	59,733,659,351	5.77	229,639
		Medical costs (total)	**73,989,945,134**	**7.15**	**284,445**
	Social welfare expenses	Disability pensions	95,238,602,667	9.20	366,133
		Public employment support services	294,382,366,078	28.45	1,131,718
		Social welfare expenses (total)	**389,620,968,745**	**37.66**	**1,497,851**
	Productivity losses	Losses due to unemployment	49,550,258,800	4.79	190,490
		Losses due to low income	315,021,989,662	30.45	1,211,064
		Absenteeism	74,227,470,116	7.17	285,359
		Presenteeism	132,263,603,686	12.78	508,471
		Productivity losses (total)	**571,063,322,264**	**55.19**	**2,195,384**
	Total burden of adult ADHD in Japan		**1,034,674,236,143**	**100.00**	**3,977,680**
Adult family members of adults with ADHD	Productivity losses	Losses due to unemployment	102,793,133,138	NA	324,892
		Losses due to low income	283,371,936,610	NA	895,636
		Absenteeism	52,671,955,750	NA	166,477
		Presenteeism	103,368,662,234	NA	326,711
		Productivity losses (total)	**542,205,687,731**	NA	**1,713,715**
Total burden on adults caused by adult ADHD in Japan			**1,576,879,923,874**	NA	NA

Total values shown may not equal the sum of individual values due to rounding

ADHD attention deficit/hyperactivity disorder, *JPY* Japanese yen, *NA* not applicable

Productivity losses in adult family members living with adults with ADHD

The number of adult family members living with adults with ADHD in Japan was estimated to be 316,392 (Additional file 1 [Table S4]). Total losses due to unemployment and low income of adult family members in Japan were estimated to be approximately 102.8 billion JPY and 283.4 billion JPY, respectively (Table 2). Total losses due to absenteeism and presenteeism in adult family members in Japan were estimated to be approximately 52.7 billion JPY and 103.4 billion JPY, respectively (Table 2). Together, the total productivity losses for adult family members of adults with ADHD in Japan were estimated to be approximately 542.2 billion JPY.

Total socioeconomic burden of adult patients with ADHD

The total socioeconomic burden of adult ADHD was estimated to be approximately 1.58 trillion JPY (Table 2; Fig. 2). Costs attributed to adults with ADHD totaled approximately 1.035 trillion JPY. Medical expenses, social welfare service expenses, and productivity loss contributed approximately 7.2%, 37.7%, and 55.2% of the socioeconomic burden of adults with ADHD, respectively (Table 2; Fig. 2). An additional cost of approximately 542 billion JPY could be attributed to productivity loss in adult family members, corresponding to 1,713,715 JPY per adult family member. The average total socioeconomic burden per adult with ADHD in Japan was estimated to be 3,977,680 JPY, which includes 54,807 JPY for outpatient medical expenses, 229,639 JPY for ADHD drug costs, 1,497,851 JPY for social welfare expenses, and 2,195,384 JPY for productivity losses (285,359 JPY and 508,471 JPY due to absenteeism and presenteeism, respectively).

Effect of psychiatric comorbidities on social welfare expenses and productivity losses

The number of Japanese adults with ADHD who had 0, 1, 2, or ≥ 3 psychiatric comorbidities was estimated to be 40,752, 87,574, 52,024, and 79,770 persons, respectively. Social welfare costs and productivity loss per adult with ADHD (base-case analysis) tended to increase with the number of psychiatric comorbidities (Fig. 3).

Scenario analysis

In scenario analyses, the number of adults with ADHD (defined as main diagnosis or subdisease) was estimated to be 272,536 and 577,062, using the top-down approach based on the "Patient Survey" conducted by the MHLW and the web-based survey, respectively. When ADHD was defined as a main diagnosis only, the estimated number of adults with ADHD was 88,976 and 90,445, using the bottom-up approach based on JMDC data and the top-down approach based on MHLW statistics, respectively.

Relative to the base case, the total estimated burden of adult ADHD in Japan increased when the number of patients with ADHD was estimated based on the web-based survey (Scenario #4: + 1,261 billion JPY; + 80.0%), when the top-down approach based on MHLW statistics was used (Scenario #2: + 30 billion JPY; + 1.9%), and when drugs used for psychiatric disorders other than ADHD were included (Scenario #5: + 115 billion JPY; + 7.3%) (Table 3). In contrast, the total estimated burden decreased when only ADHD as the main diagnosis was included (Scenario #1: – 683 billion JPY; – 43.3%), when the top-down approach based on MHLW statistics was used and only ADHD as the main diagnosis was included (Scenario #3: – 682 billion JPY; – 43.3%), when the average annual income in non-ADHD adults was derived from the web-based survey (Scenario #7: – 452 billion JPY; – 28.7%), when the unemployment rate in

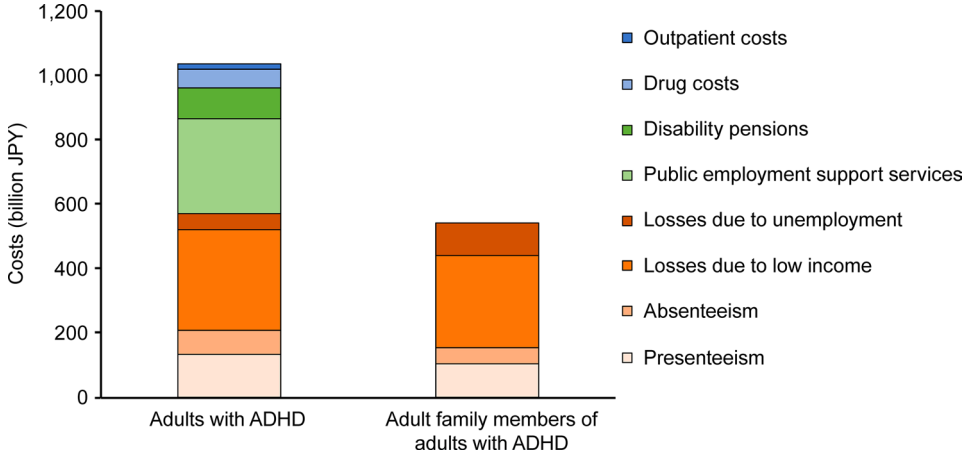


Fig. 2 Total socioeconomic burden of adult patients with ADHD and adult family members of adults with ADHD in the Japanese 2022 financial year. *ADHD* attention deficit/hyperactivity disorder, *JPY* Japanese yen

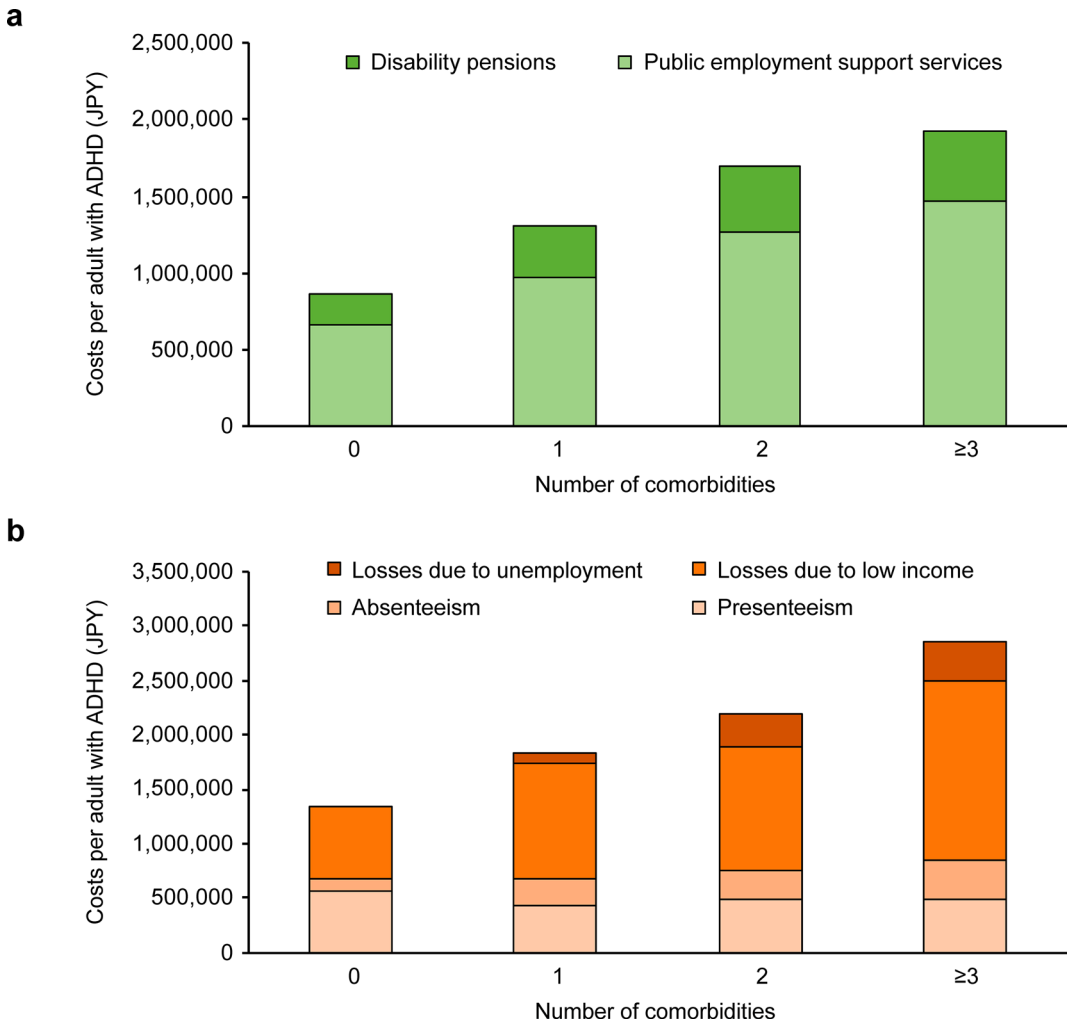


Fig. 3 (**a**) Social welfare costs and (**b**) productivity losses per adult patient with ADHD with or without comorbidities (estimated by base-case analysis) in the Japanese 2022 financial year. *ADHD* attention deficit/hyperactivity disorder, *JPY* Japanese yen

non-ADHD adults was derived from the web-based survey (Scenario #6: – 67 billion JPY; – 4.3%), and when the average annual income and unemployment rate in non-ADHD adults were derived from the web-based survey (Scenario #8: – 503 billion JPY; – 31.9%).

Relative to the base case, the total burden caused by adults with ADHD under all the different scenarios ranged from – 68.9% (– 1,086 billion JPY) to + 96.1% (+ 1,515 billion JPY) (Additional file 1 [Fig. S2]). The maximum costs occurred when (1) drugs included those for psychiatric disorders in addition to ADHD, (2) the number of patients with ADHD was estimated based on the web-based survey, and (3) the average annual income and the unemployment rate in non-ADHD adults were derived from publicly available sources. The minimum costs occurred when (1) drugs included only those for ADHD, (2) the top-down approach based on MHLW statistics was used and only ADHD as the main diagnosis was included, and (3) the average annual income and the unemployment rate in non-ADHD adults were derived from the web-based survey.

Sensitivity analysis

In the sensitivity analysis, productivity losses in adults with ADHD estimated without matching were similar to the base-case analysis (with matching for sex, age, and place of residence) (Additional file 1 [Table S5]). In contrast, matching for five variables (sex, age, and place of residence, plus educational background and occupation) reduced losses due to unemployment, losses due to low income, and total productivity losses compared with the base-case analysis. In adult family members, productivity losses were similar to the base case when estimated without matching. Analysis after matching for five variables reduced losses due to unemployment and low income, resulting in lower total productivity losses (approximately 491 billion JPY compared with 542 billion JPY in the base case).

Table 3 Scenario (sensitivity) analysis of the annual socioeconomic burden of adult ADHD in the Japanese 2022 financial year relative to the base-case analysis

Scenario	Percentage relative to base-case analysis Difference in cost from base-case analysis (JPY)					
	Medical costs	Social welfare expenses	Productivity losses	Total burden of adult ADHD population in Japan (A)	Productivity losses in adult family members (B)	Total burden of adult ADHD in Japan (A + B)
1. Estimation method for number of adults with ADHD and medical costs: bottom-up approach based on JMDC; Definition of ADHD: ADHD as main diagnosis	– 68.2% – 50,426,592,086	– 65.8% – 256,348,189,585	– 65.8% – 375,726,823,103	– 66.0% – 682,501,604,774	0% 0	– 43.3% – 682,501,604,774
2. Estimation method for number of adults with ADHD and medical costs: top-down approach based on MHLW statistics; Definition of ADHD: ADHD as main diagnosis or subdisease	– 22.0% – 16,285,280,455	4.8% 18,597,316,423	4.8% 27,257,889,471	2.9% 29,569,925,439	0% 0	1.9% 29,569,925,439
3. Estimation method for number of adults with ADHD and medical costs: top-down approach based on MHLW statistics; Definition of ADHD: ADHD as main diagnosis	– 75.3% – 55,746,065,329	– 65.2% – 254,147,846,655	– 65.2% – 372,501,803,802	– 66.0% – 682,395,715,786	0% 0	– 43.3% – 682,395,715,786
4. Estimation method for number of adults with ADHD and medical costs: bottom-up approach based on web-based survey; Definition of ADHD: ADHD as main diagnosis or subdisease	121.8% 90,152,703,332	121.8% 474,731,850,976	121.8% 695,809,439,816	121.8% 1,260,693,994,124	0% 0	80.0% 1,260,693,994,124
5. Definitions of the drugs: Medications for psychiatric disorders included	155.0% 114,671,203,371	0% 0	0% 0	11.1% 114,671,203,371	0% 0	7.3% 114,671,203,371
6. Unemployment rate in non-ADHD adults: based on web survey	0% 0	0% 0	– 1.3% – 7,298,100,133	– 0.7% – 7,298,100,133	– 11.1% – 60,031,949,951	– 4.3% – 67,330,050,085
7. Average annual income in non-ADHD adults: based on web survey	0% 0	0% 0	– 25.4% – 144,832,459,274	– 14.0% – 144,832,459,274	– 56.6% – 307,086,850,669	– 28.7% – 451,919,309,942
8. Average unemployment rate and annual income in non-ADHD adults: based on web survey	0% 0	0% 0	– 26.5% – 151,055,210,940	– 14.6% – 151,055,210,940	– 64.9% – 351,699,473,586	– 31.9% – 502,754,684,526

ADHD attention deficit/hyperactivity disorder, *JPY* Japanese yen, *MHLW* Ministry of Health, Labour and Welfare

Discussion

This is the first report to estimate the socioeconomic burden of adult ADHD in Japan from a societal perspective and the first in any Asian country. Our analysis not only included medical costs, but also social support welfare expenses and productivity losses associated with adults who have ADHD and their adult family members. The total annual costs attributable to adult ADHD in Japan were estimated to exceed 1.5 trillion JPY (approximately US$10.9 billion), with costs of nearly 4 million JPY (approximately US$29,112) per adult with ADHD.

More than half (55%) of the total socioeconomic burden for adults with ADHD was due to work-related productivity losses, while social welfare expenses and medical costs accounted for approximately 38% and 7% of overall costs, respectively. The large contribution of productivity losses is generally consistent with socioeconomic studies of adult ADHD conducted in the USA and Europe, although the specific costs vary depending on the social and medical environments in each country and the estimation methods used [27, 42, 43]. In one US analysis, 71–83% of costs were due to productivity losses (including disability services and worker's compensation, which we categorized as social welfare expenses) and income losses, with medical costs contributing 15–26% [27]. Another US analysis reported that 54.4% of costs were due to unemployment, 23.4% to productivity losses due to absenteeism and presenteeism, and 11.6%

to healthcare services [43]. Although ADHD in adults is often under-recognized, including in Japan [44], our analysis suggests that the socioeconomic burden of adult ADHD in Japan may be of a similar magnitude to the burdens reported for depression [45], anxiety [46], and schizophrenia [47], where unemployment and/or productivity losses accounted for the majority of costs. However, direct comparison of the costs of these disorders is not possible because of differences in methodology and time frame.

Productivity losses in adult family members living with ADHD adults cost approximately 1.7 million JPY per family member per year, roughly half of the total socioeconomic burden estimated for each adult with ADHD. This result is consistent with a previous study in the USA, which reported that the total burden of adult family members of adults with ADHD (US$6.5 billion) was half the total burden of adults with ADHD (US$4.6 billion for female patients and US$8.4 billion for male patients) [23]. This burden has been interpreted as a spillover effect due to the burden of living with adults with ADHD. Living with someone who has ADHD is often stressful, and the impact of ADHD on family members is wide-ranging [48]. In addition, ADHD is known to have a hereditary component [49]. In the present study, participants who self-reported ADHD were included in the ADHD adult group rather than in the family member group. However, the mean ASRS score in the family member group tended to be higher than that in the ASRS-negative non-ADHD group, suggesting that some family members had ADHD symptoms below the diagnostic threshold. In addition, adult family members had higher rates of some psychiatric comorbidities than the ASRS-negative non-ADHD control population, particularly autism spectrum disorder and depression. These comorbidities may have contributed to productivity losses in the family member population, in addition to losses directly related to living with another adult who has ADHD.

More appropriate support for adults with ADHD and their families, particularly measures aimed at facilitating more sustainable employment, may help reduce the social and economic burden of adult ADHD. Employers and society as a whole may not fully understand the characteristics of people with neurodevelopmental disorders and the environmental adjustment necessary to help them succeed [19, 21]. Indeed, people with developmental disorders including ADHD can show positive characteristics (e.g. creative thinking) in a work environment that has been appropriately adjusted for the employee [19]. Policy reform and education of employers and coworkers who do not have ADHD may be beneficial in order to foster inclusive workplaces for people with neurodevelopmental disorders. Specific examples of reasonable accommodations that will assist employees with ADHD include customization of the work environment, flexible working arrangements, structured job responsibilities, supportive technology, task breakdown, scheduling assistance, digital reminders, coaching and mentoring from supervisors and coworkers, and external assistance when required [19, 50]. Additional policies include expanding vocational rehabilitation that is tailored to people with neurodevelopmental disorders and enhancing caregiver support programs for affected families. Implementation of these and other strategies is expected to reduce rates of absenteeism and presenteeism and increase work productivity among employees with ADHD and their family members.

Scenario analyses indicated that the socioeconomic burden of adult ADHD ranged from − 68.9% to + 96.1% of the base-case analysis results. This variability in estimates was strongly attributed to the variability in the estimates of the number of adult patients with ADHD. Due to the nature of self-reporting, use of the web-based survey in the scenario analyses may have overestimated the total number of Japanese adults with ADHD; to minimize this overestimation, the base-case analysis used the JMDC database to estimate the number of adults with ADHD. In contrast, it was shown that the number of adults with ADHD may be underestimated if the analysis only considers cases where ADHD is the main diagnosis; as discussed below, this may be partly because many patients with ADHD have psychiatric comorbidities that may be recorded as the main diagnosis. As such, the base-case analysis targeted not only patients who had ADHD as the primary disease but also those who had ADHD as a secondary disease, and therefore it included all patients with a diagnosis of ADHD, regardless of the presence/absence of comorbidities. Using this method, the number of patients with ADHD was estimated to be 260,120, which is small compared with the estimated prevalence of adult ADHD in Japan previously reported in a survey conducted in a single Japanese city (1.65%) [10]. This is possibly because the previous survey estimated the true prevalence by screening a sample of the general population, whereas our study calculated the number of adults who had been diagnosed with ADHD. To avoid overestimating the number of adults with ADHD, we took a conservative approach and estimated the number of adults with an actual diagnosis of ADHD. However, evidence suggests that potentially there are adults with ADHD in Japan who are still not diagnosed with the disease [44]. If the burden of these undiagnosed individuals were included, the economic burden would be larger than our base-case estimate. Clearly, accurate numbers of adult patients with ADHD are vital for a more accurate estimation of socioeconomic burden. It is noteworthy that the base-case analysis using the JMDC database produced medical cost estimates approximately 22%

higher than those derived from the top-down approach based on MHLW statistics (Scenario #2), while the total cost estimates were in very close agreement. The MHLW statistics estimate annual medical and pharmaceutical expenditures using the National Database of Health Insurance Claims and Specific Health Checkups of Japan (NDB), which includes the entire population (including the unemployed population). These findings suggest that although the JMDC database primarily covers employees and their dependents, its estimates do not result in substantial underestimation when compared with those obtained from the NDB. In other scenario analyses, productivity losses for both adults with ADHD and family members were lower when income for the ASRS-negative non-ADHD population was based on the web survey than when based on national tax agency statistics. This result may be due to low income or under-reporting of income among ASRS-negative participants in the survey.

Psychiatric comorbidities are known to be common in adults with ADHD [14, 17, 51], as was observed in the web-based survey. In many cases, a patient may be diagnosed with another psychiatric disorder before being diagnosed with ADHD [52], including in a recent analysis of the JMDC claims database [53]. As such, it is possible that the primary disease name in the electronic records was not changed, and ADHD was not recorded as the main diagnosis in the JMDC database. To ensure that such cases were captured, our base-case analysis included patients with ADHD either as the main diagnosis or as a subdisease. The observation that the base-case estimates were very similar to scenario analyses that used government statistics (i.e. top-down approach, not limited to ADHD as the main diagnosis) supports the validity of the estimates from the base-case analyses. In addition, only the cost of ADHD medications was calculated as the drug cost in the base-case analysis. We took this approach because of the difficulty of determining whether medications commonly used for other psychiatric comorbidities were being used to treat psychiatric disorders caused by ADHD [54]. In the scenario analyses, estimates that included medication costs for psychiatric comorbidities related to ADHD resulted in a 155% increase in medical costs and a 11% increase in total burden for adult patients with ADHD. In addition, in the estimation limited to patients whose primary diagnosis was ADHD (Scenario #1), medical and total costs for the adult ADHD population were approximately 66–68% lower compared with the base-case analysis, in which ADHD was either the main diagnosis or a subdisease. This suggests that many adults with ADHD have another main diagnosis, most likely comorbid psychiatric disorders. Thus, the base-case analysis likely underestimates drug costs associated with ADHD. Psychiatric comorbidities are also likely to contribute to productivity losses [55], and our subgroup analysis results suggest that productivity losses and social expenses increase with the number of comorbidities. Prevention of secondary comorbidities in adults with ADHD may reduce both the disease burden for individual patients and the overall socioeconomic burden.

In the web-based survey using Macromill panels, the diagnosis of ADHD was based on self-report by participants. We also conducted a web-based survey in patients who were prescribed ADHD drugs and visited Nihon Chouzai pharmacies; because they were prescribed ADHD-specific medication, these patients were considered likely to have a true ADHD diagnosis. Comparison of the Macromill panels and the Nihon Chouzai panel confirmed that there were no major differences in most parameters between the two survey samples. Because the values for the unemployment rate, income, absenteeism, and presenteeism were similar, we considered the effect of self-reporting in the Macromill panel to be relatively small. However, it should be noted that the use of social welfare services was higher among participants in the Macromill survey than those in the Nihon Chouzai survey, which may suggest the possibility of some overestimation of social services costs in the analysis. Sensitivity analyses identified educational background and occupation as potential confounders in the estimation of productivity losses. This may suggest that productivity loss in adults with ADHD and adult family members in Japan is at least partly related to educational background or occupation.

This study represents a comprehensive analysis of the socioeconomic burden of adult ADHD in Japan using data from a large claims database, a prospective, nationwide web-based survey, and government statistics. Our estimates include the cost of Japan's social security services that support patients with ADHD and productivity losses in adult family members living with an adult with ADHD. In addition to nationwide estimates, we report losses per patient with ADHD or family member, which allows comparison of losses between Japan and other countries with different numbers of patients with ADHD. We found by subgroup analysis that social welfare expenditure and productivity loss tended to increase according to the number of psychiatric comorbidities. Multiple scenario analyses were conducted to evaluate the effect of different methods on estimated costs, including top-down and bottom-up approaches, and sensitivity analyses that used different background factors for matching identified potential confounders for productivity loss in adults with ADHD and their families.

Despite these advantages, our study does have some limitations. For example, the JMDC is a database of employees' insurance associations, which are not evenly distributed throughout Japan, and potential regional differences cannot be ascertained. In addition, because the

JMDC database consists of employees and their dependents, data for unemployed people and people older than 65 years are limited, which could result in an underestimation of the number of adults with ADHD and the associated costs. However, the number of adults with ADHD estimated in our base-case analysis (260,120) was similar to the results of a top-down approach using the Patient Survey conducted by the MHLW (272,536), suggesting that there is no substantial underestimation. The Patient Survey is used to estimate the total number of patients and treatment rates nationwide and is conducted on patients (including unemployed patients) who visited medical institutions across Japan during the survey period; institutions are selected using a stratified random sampling method. In addition, the large costs associated with decreased work productivity would not be affected by the exclusion of non-working people, and the findings from Scenario #2 (described above) support this suggestion. More precise estimation of the medical costs of adult ADHD using other databases, such as the NDB, would be valuable. Limitations of the web-based survey include the possibility of recall bias (e.g. when asked to recall how work was affected over the previous 7 days) and sampling bias (e.g. only people with access to a computer or smartphone could be included). As noted, adults with ADHD often have other psychiatric disorders, so losses attributed specifically to ADHD may be overestimated. To minimize the effect of this limitation, we focused on outpatient medical costs without including inpatient medical costs (which are rare in Japan), and we included only drugs specific for ADHD treatment in the base-case analysis. However, we did not include non-medical costs such as transportation costs in our analysis. Our scenario analyses indicated that the number of adults with ADHD varied with the estimation method used and greatly affected the calculated total losses. Greater certainty in the number of adults with ADHD would improve the robustness of future cost analyses. In addition, because ADHD is often diagnosed in childhood or adolescence, the estimated costs in adults may not include costs associated with the diagnosis itself or early intervention. Further study to estimate the socioeconomic burden of childhood ADHD using appropriate methods is warranted. Finally, losses due to lawsuits, premature deaths, or accidents, as well as any impact of dropping out of school, were not included in the study because appropriate data sources could not be identified.

Conclusions

This study, the first estimate of the annual socioeconomic burden of adult ADHD in Japan, underscores the necessity of providing appropriate support to adult patients with ADHD and their families and the prevention of comorbidities. In particular, measures to support participation in the workplace to improve patients' quality of life and reduce the social and economic burden should be explored. Furthermore, it is hoped that estimating the annual socioeconomic burden of adult ADHD will help clarify policy priorities in Japan.

Abbreviations

ADHD	Attention-deficit/hyperactivity disorder
ASRS	Adult ADHD Self-Report Scale
JPY	Japanese yen
MHLW	Ministry of Health, Labour and Welfare
SD	Standard deviation

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12888-026-07953-x.

Supplementary Material 1

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Author contributions

All authors participated in the study design, in the interpretation of study results, and in the drafting, critical revision, and approval of the final version of the manuscript. AI contributed to the study design and interpretation as a health technology assessment expert. NN was involved in the original conception of the study. NT contributed to the study design and interpretation as a medical expert.

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Data availability

A portion of the data that support the findings of this study is available from JMDC Inc. but is used under license for the current study; therefore, restrictions apply and the data are not publicly available. For inquiries about access to the data set used in this study, please contact JMDC (https://www.jmdc.co.jp/en/). Many of the data from the web-based survey generated or analyzed during this study are included in this published article and its supplementary information files. Raw data derived from the web-based survey are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Non-Profit Organization MINS Institutional Review Board (ID: 230222) and registered at the University hospital Medical Information Network Center (UMIN000051654) on July 20, 2023. Participants in the web-based surveys provided informed consent before starting the survey. As data from the JMDC claims database were retrospective and anonymous, informed consent was not required.

Consent for publication

Not applicable.

Competing interests
AI reports that his affiliation, the Department of Health Economics and Outcomes Research, Graduate School of Pharmaceutical Sciences, The University of Tokyo, is an endowment department and is supported with an unrestricted grant from Takeda Pharmaceutical Company Limited.NN is a former employee of Takeda Pharmaceutical Company Limited. YK is an employee of Takeda Pharmaceutical Company Limited. NT belongs to an endowed chair founded by the Toyama prefecture in Japan and reports receiving personal fees from Mitsubishi Tanabe Pharma Corporation, Mochida Pharmaceutical Co., Ltd., Otsuka Pharmaceutical Co., Ltd., Shionogi & Co., Ltd., Takeda Pharmaceutical Company Limited, Yoshitomi Pharmaceutical Industries Ltd., and Nobelpharma Co., Ltd.

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References

1. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision (DSM-5-TR). Washington, DC: American Psychiatric Association Publishing; 2022.
2. Faraone SV, Asherson P, Banaschewski T, Biederman J, Buitelaar JK, Ramos-Quiroga JA, et al. Attention-deficit/hyperactivity disorder. Nat Rev Dis Primers. 2015;1:15020.
3. Posner J, Polanczyk GV, Sonuga-Barke E. Attention-deficit hyperactivity disorder. Lancet. 2020;395(10222):450–62.
4. Thomas R, Sanders S, Doust J, Beller E, Glasziou P. Prevalence of attention-deficit/hyperactivity disorder: a systematic review and meta-analysis. Pediatrics. 2015;135(4):e994–1001.
5. Simon V, Czobor P, Bálint S, Mészáros A, Bitter I. Prevalence and correlates of adult attention-deficit hyperactivity disorder: meta-analysis. Br J Psychiatry. 2009;194(3):204–11.
6. Ayano G, Tsegay L, Gizachew Y, Necho M, Yohannes K, Abraha M, et al. Prevalence of attention deficit hyperactivity disorder in adults: umbrella review of evidence generated across the globe. Psychiatry Res. 2023;328:115449.
7. Willcutt EG. The prevalence of DSM-IV attention-deficit/hyperactivity disorder: a meta-analytic review. Neurotherapeutics. 2012;9(3):490–9.
8. Kanbayashi Y, Nakata Y, Fujii K, Kita M, Wada K. ADHD-related behavior among non-referred children: parents' ratings of DSM-III-R symptoms. Child Psychiatry Hum Dev. 1994;25(1):13–29.
9. Nomura K, Okada K, Noujima Y, Kojima S, Mori Y, Amano M, et al. A clinical study of attention-deficit/hyperactivity disorder in preschool children–prevalence and differential diagnoses. Brain Dev. 2014;36(9):778–85.
10. Nakamura S, Ohnishi M, Uchiyama S. Epidemiological survey of adult attention deficit hyperactivity disorder (ADHD) in Japan [in Japanese]. Jpn J Psychiatr Treat. 2013;28:155–62.
11. Sasayama D, Kuge R, Toibana Y, Honda H. Trends in diagnosed attention-deficit/hyperactivity disorder among children, adolescents, and adults in Japan from April 2010 to March 2020. JAMA Netw Open. 2022;5(9):e2234179.
12. Faraone SV, Biederman J, Mick E. The age-dependent decline of attention deficit hyperactivity disorder: a meta-analysis of follow-up studies. Psychol Med. 2006;36(2):159–65.
13. Lara C, Fayyad J, de Graaf R, Kessler RC, Aguilar-Gaxiola S, Angermeyer M, et al. Childhood predictors of adult attention-deficit/hyperactivity disorder: results from the World Health Organization World Mental Health Survey Initiative. Biol Psychiatry. 2009;65(1):46–54.
14. Fayyad J, Sampson NA, Hwang I, Adamowski T, Aguilar-Gaxiola S, Al-Hamzawi A, et al. The descriptive epidemiology of DSM-IV Adult ADHD in the World Health Organization World Mental Health Surveys. Atten Defic Hyperact Disord. 2017;9(1):47–65.
15. Able SL, Haynes V, Hong J. Diagnosis, treatment, and burden of illness among adults with attention-deficit/hyperactivity disorder in Europe. Pragmat Obs Res. 2014;5:21–33.
16. Kessler RC, Adler L, Ames M, Barkley RA, Birnbaum H, Greenberg P, et al. The prevalence and effects of adult attention deficit/hyperactivity disorder on work performance in a nationally representative sample of workers. J Occup Environ Med. 2005;47(6):565–72.
17. Kirino E, Imagawa H, Goto T, Montgomery W. Sociodemographics, comorbidities, healthcare utilization and work productivity in Japanese patients with adult ADHD. PLoS One. 2015;10(7):e0132233.
18. Adamou M, Arif M, Asherson P, Aw TC, Bolea B, Coghill D, et al. Occupational issues of adults with ADHD. BMC Psychiatry. 2013;13:59.
19. Sarkis E. Addressing attention-deficit/hyperactivity disorder in the workplace. Postgrad Med. 2014;126(5):25–30.
20. Corrigan PW, Watson AC. Understanding the impact of stigma on people with mental illness. World Psychiatry. 2002;1(1):16–20.
21. Doyle N. Neurodiversity at work: a biopsychosocial model and the impact on working adults. Br Med Bull. 2020;135(1):108–25.
22. Ministry of Health, Labour and Welfare. Health care and welfare measures for persons with disabilities. Annual Health, Labour and Welfare Report 2021. https://www.mhlw.go.jp/english/wp/wp-hw14/dl/09e.pdf. Accessed 29 Apr 2024.
23. Birnbaum HG, Kessler RC, Lowe SW, Secnik K, Greenberg PE, Leong SA, et al. Costs of attention deficit-hyperactivity disorder (ADHD) in the US: excess costs of persons with ADHD and their family members in 2000. Curr Med Res Opin. 2005;21(2):195–205.
24. LeFevre-Levy R, Melson-Silimon A, Harmata R, Hulett AL, Carter NT. Neurodiversity in the workplace: considering neuroatypicality as a form of diversity. Ind Organ Psychol. 2023;16(1):1–19.
25. Smith T, Kirby A. Neurodiversity at Work: Drive Innovation, Performance and Productivity With a Neurodiverse Workforce. London, UK: Kogan Page Limited; 2021.
26. Chhibber A, Watanabe AH, Chaisai C, Veettil SK, Chaiyakunapruk N. Global economic burden of attention-deficit/hyperactivity disorder: a systematic review. Pharmacoeconomics. 2021;39(4):399–420.
27. Doshi JA, Hodgkins P, Kahle J, Sikirica V, Cangelosi MJ, Setyawan J, et al. Economic impact of childhood and adult attention-deficit/hyperactivity disorder in the United States. J Am Acad Child Adolesc Psychiatry. 2012;51(10):990–e1002.e2.
28. Le HH, Hodgkins P, Postma MJ, Kahle J, Sikirica V, Setyawan J, et al. Economic impact of childhood/adolescent ADHD in a European setting: the Netherlands as a reference case. Eur Child Adolesc Psychiatry. 2014;23(7):587–98.
29. Takeda T, Tsuji Y, Kurita H. Psychometric properties of the Japanese version of the Adult Attention-deficit hyperactivity disorder (ADHD) Self-Report Scale (ASRS-J) and its short scale in accordance with DSM-5 diagnostic criteria. Res Dev Disabil. 2017;63:59–66.
30. Tsujii N. Attention Deficit Hyperactivity Disorder Assessment Scales and Tests - ASRS-v1.1, ADHD-RS-5, Conners 3 Japanese version, CAARS Japanese version, WURS, CAADID Japanese version, DIVA-5 [in Japanese]. Seisin Igaku. 2024;66(5):509–13.
31. Kessler RC, Adler L, Ames M, Demler O, Faraone S, Hiripi E, et al. The World Health Organization Adult ADHD Self-Report Scale (ASRS): a short screening scale for use in the general population. Psychol Med. 2005;35(2):245–56.
32. World Health Organization, Adult ADHD Working Group. Adult ADHD Self-Report Scale version 1.1 (ASRS-v1.1) [Japanese translation]. Accessed 1 Jun 2023. https://www.hcp.med.harvard.edu/ncs/ftpdir/adhd/adhd/Old%20Versions/18Q_Japanese_final_updated.pdf
33. Wakabayashi K, Takada A, Matsumoto T, Yasuhara M, Kijima Y. Furthering opportunities for people with developmental disorders in a digital society, and the economic impact. Accessed 30 Apr 2024. https://www.nri.com/en/journal/2021/0507
34. Team for Revision of Remuneration for Welfare Services for Persons with Disabilities. Remuneration and standards for employment transition support and employment retention support: discussion points [in Japanese]. https://www.mhlw.go.jp/content/12401000/000674638.pdf. Accessed 1 May 2024.
35. Team for Revision of Remuneration for Welfare Services for Persons with Disabilities. Remuneration and standards for continuous employment support: discussion points [in Japanese]. https://www.mhlw.go.jp/content/12401000/000674639.pdf. Accessed 1 May 2024.
36. Reilly MC, Zbrozek AS, Dukes EM. The validity and reproducibility of a work productivity and activity impairment instrument. Pharmacoeconomics. 1993;4(5):353–65.
37. Ministry of Internal Affairs and Communications. Table 1: age (year), population, and population and sex ratio by sex - total population, Japanese

population (as of October 1, 2022) [in Japanese]. https://www.stat.go.jp/data/jinsui/2022np/zuhyou/05k2022-1.xlsx. Accessed 1 May 2024.
38. Ministry of Internal Affairs and Communications. National Census 2020 - basic aggregate results of employment status [in Japanese]. https://www.stat.go.jp/data/kokusei/2020/kekka/pdf/outline_02.pdf. Accessed 1 May 2024.
39. Ministry of Internal Affairs and Communications. Labour Force Survey (Basic Tabulation) 2022 (Reiwa 4) - summary of average results [in Japanese]. https://www.stat.go.jp/data/roudou/sokuhou/nen/ft/pdf/index1.pdf. Accessed 1 May 2024.
40. National Tax Agency. 2021 survey of private sector salaries [in Japanese]. https://www.nta.go.jp/publication/statistics/kokuzeicho/minkan/gaiyou/2021.htm#a-01. Accessed 1 May 2024.
41. Ministry of Health, Labour and Welfare. Monthly Labor Survey: final results for fiscal year 2022 [in Japanese]. https://www.mhlw.go.jp/toukei/itiran/roudou/monthly/r04/22fr/dl/pdf22fr.pdf. Accessed 1 May 2024.
42. Daley D, Jacobsen RH, Lange AM, Sørensen A, Walldorf J. The economic burden of adult attention deficit hyperactivity disorder: a sibling comparison cost analysis. Eur Psychiatry. 2019;61:41–8.
43. Schein J, Adler LA, Childress A, Gagnon-Sanschagrin P, Davidson M, Kinkead F, et al. Economic burden of attention-deficit/hyperactivity disorder among adults in the United States: a societal perspective. J Manag Care Spec Pharm. 2022;28(2):168–79.
44. Naya N, Tsuji T, Nishigaki N, Sakai C, Chen Y, Jung S, et al. The burden of undiagnosed adults with attention-deficit/hyperactivity disorder symptoms in Japan: a cross-sectional study. Cureus. 2021;13(11):e19615.
45. Sado M, Yamauchi K, Kawakami N, Ono Y, Furukawa TA, Tsuchiya M, et al. Cost of depression among adults in Japan in 2005. Psychiatry Clin Neurosci. 2011;65(5):442–50.
46. Sado M, Takechi S, Inagaki A, Fujisawa D, Koreki A, Mimura M, et al. Cost of anxiety disorders in Japan in 2008: a prevalence-based approach. BMC Psychiatry. 2013;13(1):338.
47. Sado M, Inagaki A, Koreki A, Knapp M, Kissane LA, Mimura M, et al. The cost of schizophrenia in Japan. Neuropsychiatr Dis Treat. 2013;9:787–98.
48. Harpin VA. The effect of ADHD on the life of an individual, their family, and community from preschool to adult life. Arch Dis Child. 2005;90 Suppl 1(Suppl 1):i2–7.
49. Franke B, Faraone SV, Asherson P, Buitelaar J, Bau CHD, Ramos-Quiroga JA, et al. The genetics of attention deficit/hyperactivity disorder in adults, a review. Mol Psychiatry. 2012;17(10):960–87.
50. Schreuer N, Dorot R. Experiences of employed women with attention deficit hyperactive disorder: a phenomenological study. Work. 2017;56(3):429–41.
51. Chen Q, Hartman CA, Haavik J, Harro J, Klungsøyr K, Hegvik T-A, et al. Common psychiatric and metabolic comorbidity of adult attention-deficit/hyperactivity disorder: a population-based cross-sectional study. PLoS One. 2018;13(9):e0204516.
52. Piñeiro-Dieguez B, Balanzá-Martínez V, García-García P, Soler-López B, CAT Study Group. Psychiatric comorbidity at the time of diagnosis in adults with ADHD: the CAT study. J Atten Disord. 2016;20(12):1066–75.
53. Okada T, Sotodate T, Ogasawara-Shimizu M, Nishigaki N. Psychiatric comorbidities of attention deficit/hyperactivity disorder in Japan: a nationwide population-based study. Front Psychiatry. 2024;15:1359872.
54. Wong J, Abrahamowicz M, Buckeridge DL, Tamblyn R. Assessing the accuracy of using diagnostic codes from administrative data to infer antidepressant treatment indications: a validation study. Pharmacoepidemiol Drug Saf. 2018;27(10):1101–11.
55. de Oliveira C, Saka M, Bone L, Jacobs R. The role of mental health on workplace productivity: a critical review of the literature. Appl Health Econ Health Policy. 2023;21(2):167–93.

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